



Realized skewness of oil price returns and the short-term predictability for exchange rate

Yaojie Zhang, Yuqing Feng & Chaojian Wei

To cite this article: Yaojie Zhang, Yuqing Feng & Chaojian Wei (2025) Realized skewness of oil price returns and the short-term predictability for exchange rate, Quantitative Finance, 25:7, 1031-1046, DOI: [10.1080/14697688.2025.2522177](https://doi.org/10.1080/14697688.2025.2522177)

To link to this article: <https://doi.org/10.1080/14697688.2025.2522177>



Published online: 15 Jul 2025.



Submit your article to this journal [↗](#)



Article views: 268



View related articles [↗](#)



View Crossmark data [↗](#)

Realized skewness of oil price returns and the short-term predictability for exchange rate

YAOJIE ZHANG , YUQING FENG* and CHAOJIAN WEI

School of Economics and Management, Nanjing University of Science and Technology, Nanjing, People's Republic of China

(Received 8 January 2025; accepted 16 June 2025)

When predicting exchange rate returns, existing models find it difficult to defeat naive random walk models. We propose the realized skewness of oil returns to predict exchange rates for nine countries. The results show that it is statistically and economically more effective than the random walk model. Moreover, realized skewness can offer complementary predictive information about what is previously known about oil prices and oil volatility. It is noteworthy that the superior predictability of realized skewness is linked to inflation rates, behavioral bias, and risk transmission channels. Furthermore, the predictive power of realized skewness for exchange rates stems from the jump and drift components.

Keywords: Realized skewness; Exchange rate predictability; Inflation; Behavioral bias; Risk transmission

JEL Classifications: C22, C53, G11

1. Introduction

The foreign exchange market is essential for expanding trade between nations, promoting international capital flows, and providing investment opportunities. Therefore, it is of great significance to achieve accurate forecasts of exchange rates to maintain global economic prosperity. Meese and Rogoff (1983a) and Meese and Rogoff (1983b) argue that the random walk model of exchange rates cannot usually be defeated by the traditional economic models, especially at a short horizon. To address this problem, numerous studies have sought out new predictive variables or alternative predictive methods to improve the exchange rate forecasts (see, e.g. Li *et al.* 2015; Salisu *et al.* 2019; Liu *et al.* 2020; Ren *et al.* 2021; Iyke *et al.* 2022).

Crude oil has long been employed to predict exchange rates because of its strong linkage with the real economy. As a major driver of economic conditions, oil prices embed forward-looking information about potential economic changes. When market participants anticipate such changes based on oil price movements, these expectations are often reflected in exchange rates. However, the majority of prior studies have focused on the predictive power of fundamental factors, such as oil prices for exchange rate. Yet, there remains no consensus regarding their predictability (Chen and Chen 2007; Lizardo and Mollick 2010; Ferraro *et al.* 2015;

Kohlscheen *et al.* 2017; Han *et al.* 2024). Gao *et al.* (2022) emphasize that information on crude oil volatility serves as a more reliable predictor of economic growth compared to information on stock market volatility and other uncertainty-related variables. Realized skewness, as a measure of tail risk, can reflect the risk of an upward or downward trend in oil prices. Set against this background, we provide a potential way to predict exchange rate returns by introducing the realized skewness of oil price returns.

Our approach is to build monthly realized skewness of oil returns from January 1985 to December 2022. Taking the lagged realized skewness as a predictor, the empirical results show significant predictive power for 8 of the 9 developed countries' currencies (excluding JPY), with the largest in- and out-of-sample R^2 reaching 3.483% and 3.386% for NOK, respectively. There are two reasons for the poor performance of JPY predictions. First, the Bank of Japan's regular interventions in currency markets significantly influence yen exchange rate patterns. Additionally, the yen's reputation as a safe-haven currency contributes to the unpredictability of capital movements, especially amidst market turbulence. Consequently, forecasting the JPY's returns poses a considerable challenge.

As for the economic significance, compared to the random walk with drift model, the predictive model with realized skewness can produce higher annualized average returns, certainty equivalent return (CER), Sharpe ratios, and a lower

*Corresponding author. Email: fengyuqing@njust.edu.cn

maximum drawdown in 7 of 9 countries' currencies. The predictability of realized skewness is not driven by common predictors such as oil prices and oil volatility. Furthermore, it can offer additional information exceeding what is currently present in the common predictors. In the standard high-frequency econometric framework, realized skewness asymptotically depends only on jumps, but in finite samples, it is also influenced by drift. We examine economic explanations for realized skewness's predictive power, finding it driven by both jump and drift components. This is consistent with the theory of Bandi *et al.* (2023). Our results are robust to alternative currencies, alternative window sizes, and alternative estimation windows for half of the currencies, especially the commodity currencies.

More importantly, we discover the predictability of realized skewness for exchange rate returns mainly through inflation rate, behavioral bias, and risk transmission channels. First, the purchasing power parity (PPP) concept posits that exchange rates will align to reflect relative inflation rates between two nations, thereby equalizing their purchasing power. As one of the important commodities, oil price increases have historically preceded high inflation and low economic growth (see, e.g. Barsky and Kilian 2004; Chen 2009; Choi *et al.* 2018; Kilian and Zhou 2022). Therefore, we consider the inflation rate as one of the channels through which the realized skewness provides predictive power. Second, an investor's preference toward skewness reflects their spirit of gambling. Assets with right-skewed tend to be overpriced (see, e.g. Mitton and Vorkink 2007; Bali *et al.* 2011; Conrad *et al.* 2014), particularly amid high investor sentiment (Stambaugh *et al.* 2012). This behavioral bias can impact exchange rate movements in the foreign exchange (FX) market. Finally, according to Brunnermeier *et al.* (2008), an effective indicator of global risk appetite is the CBOE Volatility Index (VIX), where a higher VIX indicates higher returns in high-rate currencies and lower returns in low-rate currencies. The empirical results confirm that the realized skewness of oil price returns significantly negatively predicts inflation, investor sentiment, and VIX.

Our paper is closely related to the literature on modeling realized skewness. The simplest approach is to apply standardized estimates to historical returns. However, there are significant estimation errors in this estimator (Kim and White 2004). Neuberger (2012) provides an unbiased estimator for long horizon skewness by calculating third moments and incorporating implied variance, which reveals that long horizon skewness originates from both short horizon skewness components and the correlation between returns and volatility innovations (leverage effects). However, Amaya *et al.* (2015) demonstrate that realized third moments asymptotically separate jump contributions from continuous variation, specifically capturing jump-induced skewness independent of leverage effects. Neuberger and Payne (2021) empirically reveal that the long horizon skewness is driven by the leverage effect (covariance between variance and lagged returns), rather than the short horizon skewness (which decays with $T^{-1/2}$). Kelly and Jiang (2014) take an alternative approach. They use information from the cross-section of stock prices and make assumptions about the commonality of events in the tails. Liu *et al.* (2014) enhance the robustness of realized

skewness estimation by considering microstructure noise and jumps, which are common in high-frequency environments. Fukasawa and Matsushita (2021) develop a novel approach using the complete Bell polynomials to construct realized cumulants with aggregation properties, enabling efficient estimation of martingale higher-order moments. Our formula follows Amaya *et al.* (2015) on the calculation of skewness but differs slightly from theirs. They use intraday returns to build realized skewness. We compute monthly realized skewness using the trading day data to mitigate the impact of noisy information embedded in the high-frequency oil price data (Bandi and Russell 2008).

Our paper is closely aligned with the existing literature on the predictive power of realized skewness in asset returns. Pioneering work by Kraus and Litzenberger (1976) integrated the concept of skewness into the Capital Asset Pricing Model (CAPM), while Harvey and Siddique (2000) emphasize the necessity of accounting for skewness risk in asset return expectations. The preceding studies confirm the predictive power of skewness in forecasting expected returns across various markets, including the stock market (see, e.g. Conrad *et al.* 2013; Amaya *et al.* 2015; Jondeau *et al.* 2019), the FX market (Jiang *et al.* 2019c; Liu *et al.* 2022), and the commodity futures market (Fernandez-Perez *et al.* 2018). In the FX market, Jiang *et al.* (2019c) demonstrate that currency skewness, measured as the difference between upside and downside variance risk premia, positively predicts cross-sectional currency excess returns. Liu *et al.* (2022) further prove this positive relationship in a time-series analysis of the CNY-CNH spread, though their study omits economic testing. The positive relationship is consistent with undiversified investors demanding compensation for skewness risk (Clark and West 2007; Mitton and Vorkink 2007). Such investors, driven by the pursuit of high skewness exposure, often sacrifice portfolio variance efficiency. Our paper contributes to this body of research by introducing crude oil skewness as a novel predictor for FX time-series returns and uncovering a negative time-series relationship. We attribute this relationship to speculative overvaluation dynamics, where investors are drawn to assets with positive skewness in pursuit of lottery-like returns. Such behavior leads to overvaluation of these assets and subsequently lower expected returns. This explanation aligns with the mixed empirical evidence on the relationship between skewness and expected asset returns (see, e.g. Cremers and Weinbaum 2010; Amaya *et al.* 2015; Jiang *et al.* 2019b), providing a plausible rationale for our findings.

Additionally, our paper has a tight relationship to the literature on oil information to forecast exchange rate returns. Yin *et al.* (2022) find that the oil's volatility risk premium (VRP) can significantly predict the main commodity-related currency returns. Breen and Hu (2021) show the CBOE Crude Oil ETF Volatility Index (OVX) is a reliable forecaster for the Canadian dollar on a daily frequency. Musa *et al.* (2022) link oil price uncertainty, as defined by Nguyen *et al.* (2022), to a depreciation in the currencies of major oil-exporting nations. Additionally, Śmiech *et al.* (2021) use the oil price uncertainty developed by Yin and Feng (2019) to observe that such uncertainty shocks lead to immediate depreciation in the currencies of oil-exporting nations such as Canada, Mexico, Norway, and Russia. Interestingly, this reaction persists over time only in

developing countries like Mexico and Russia. By contrast, our main contribution lies in introducing realized skewness of oil price returns as a novel indicator, which provides new insight into the interaction between currency markets and the oil sector. We also explore the predictive power of realized skewness on exchange rate returns through three potential channels: inflation, investor behavior, and risk transfer mechanisms. This approach provides a comprehensive analysis that goes beyond statistical and economic performance.

The paper is structured as follows. Section 2 introduces the variables and methods. Section 3 portrays the source of the data. Section 4 shows the empirical findings. Section 5 gives the economic explanation behind the predictive power. Section 6 conducts a series of robustness tests. Finally, Section 7 concludes.

2. Methodology

2.1. Exchange rate return

We use the first-order log differences of the exchange rate to calculate monthly returns,

$$r_{i,t+1} = \log(S_{i,t+1}) - \log(S_{i,t}), \quad (1)$$

where $S_{i,t}$ denotes the nominal exchange rate of the i -th currency in the t -th month, expressed as the number of domestic currency units per US dollar.

2.2. Realized skewness

Merton (1980) suggests that increasing the sampling frequency enables the measurement of volatility with arbitrary precision. Following this idea, the realized volatility literature (Andersen and Bollerslev 1998; Andersen *et al.* 2003) proposes a measure of time-varying volatility. Specifically, it has facilitated the computation of daily realized volatility using intraday squared returns. Building on this foundation, Amaya *et al.* (2015) further extended the concept of realized volatility by calculating realized skewness from the intraday cubic returns. French *et al.* (1987), Schwert (1989), and Paye (2012) construct realized volatility at the monthly frequency by summing the squared daily returns. Similarly, we construct the monthly realized skewness of oil price returns by adding together the cubic daily returns[‡] to mitigate the impact of noisy information embedded in the high-frequency oil price data (Bandi and Russell 2008). Firstly, we create the realized

volatility (RV) of oil price on a monthly frequency,

$$RV_t = \sum_{i=1}^{N_t} r_{i,t}^2, \quad (2)$$

where $r_{i,t}$ denotes the logarithmic return of the i -th trading day for oil futures during month t , and N_t denotes the total count of trading days in the t -th month.

Then, using daily returns scaled by the monthly RV, we compute monthly realized skewness (RSk) of oil price returns as follows[‡]:

$$RSk_t = \frac{\sqrt{N_t} \sum_{i=1}^{N_t} r_{i,t}^3}{RV_t^{1.5}}. \quad (3)$$

Theoretically, the values of realized skewness reflect the asymmetry of asset returns. In the literature on asset pricing and risk management, negative skewness is often closely associated with tail risk in the market, and thus skewness can be regarded as a measure of tail risk (Kelly and Jiang 2014; Amaya *et al.* 2015; Bollerslev *et al.* 2015). There is a great downside risk for asset returns when looking at the negative values.

2.3. Predictive model

We employ the univariate regression model to analyze the in-sample performance, which can be expressed as:

$$r_{i,t+1} = \alpha_i + \beta_i RSk_t + \varepsilon_{i,t+1}, \quad (4)$$

where $r_{i,t+1}$ stands for the exchange rate returns of the i -th currency from month t to $t + 1$, RSk_t denotes realized skewness of oil price returns in the t -th month, and $\varepsilon_{i,t+1}$ is the disturbance term. The t -statistic is used to test β_i with the null hypothesis unpredictable ($H_0: \beta_i = 0$). We compute the standard errors using the Newey and West (1987) approach to take into consideration the serial correlation.

In line with prior research on predictability (see, e.g. Rapach *et al.* 2010; Neely *et al.* 2014; Huang *et al.* 2015; Rapach *et al.* 2016; Jiang *et al.* 2019a; Hong and Wei 2025), we construct out-of-sample forecasts of exchange rate returns relying on an expanding estimation window.[§] Specifically,

[‡] We use daily frequency data rather than high-frequency intraday data. Stochastic price processes such as Brownian motion and jump-diffusion models are commonly applied in high-frequency analysis, as they capture the micro-dynamic features of prices. These processes provide a mathematical framework to derive the limiting characteristics of price changes as sampling frequency approaches infinity, which is crucial for understanding market microstructure. For example, in jump-diffusion models, price changes may include jumps, which are more evident at higher sampling frequencies. However, daily data smooth out these details through temporal aggregation, reducing the significance of defining stochastic processes at this frequency.

[‡] The numerator of skewness is the sum of the third powers of returns, which captures both the direction and the magnitude of distributional asymmetry. Unlike the second power used in volatility calculations, the third power amplifies the influence of extreme values, particularly enhancing sensitivity to tail events. The denominator is the 1.5th power of RV, which standardizes the distribution, thereby rendering skewness invariant to the scale of returns per unit time and enabling comparisons across different assets and time intervals. This functional form can capture tail risk more sensitively, particularly the possibility of extreme negative returns, through the third power of returns and normalization.

[§] The rolling window is favored for daily return forecasts, whereas the expanding window is commonly used for monthly return predictions. This stems from the fact that daily data series are more susceptible to structural breaks, and the rolling window can help alleviate this problem. Conversely, monthly data series are less prone to such breaks, making the expanding window more suitable. Additionally, the model parameters with expanding window are always re-estimated using all the previous observations, which is beneficial for monthly predictions where the number of observations is typically limited.

the regression model is constructed utilizing all available historical data up to period t . Subsequently, these estimated parameters and predictors from period t are employed to generate a return forecast for period $t + 1$. This process is then iteratively expanded by incorporating additional observations, and the cycle is repeated to produce out-of-sample predictions until the end of the sample. The entire T -month sample is divided into two segments: an initial in-sample period consisting of the first m observations, and an out-of-sample period containing the remaining $T - m$ observations. The first out-of-sample forecast is provided by,

$$\hat{r}_{i,m+1} = \hat{\alpha}_{i,m} + \hat{\beta}_{i,m}RSk_m, \quad (5)$$

where $\hat{\alpha}_{i,m}$ and $\hat{\beta}_{i,m}$ are the estimations of α_i and β_i by the ordinary least squares (OLS) approach, which is obtained by regressing $\{r_t\}_{t=2}^m$ on $\{RSk_t\}_{t=1}^{m-1}$ and a constant. The second out-of-sample forecast is provided by

$$\hat{r}_{i,m+2} = \hat{\alpha}_{i,m+1} + \hat{\beta}_{i,m+1}RSk_{m+1}, \quad (6)$$

where $\hat{\alpha}_{i,m+1}$ and $\hat{\beta}_{i,m+1}$ are obtained by regressing $\{r_t\}_{t=2}^{m+1}$ on $\{RSk_t\}_{t=1}^m$ and a constant. By following this pattern until the conclusion of the out-of-sample period, we can obtain $T-m$ predictions of exchange rate returns.

We use the Random Walk (RW) model as the benchmark model to evaluate the performance of our proposed predictive model. This choice is largely rooted in the seminal work of Meese and Rogoff (1983a) and Meese and Rogoff (1983b). Their work has robustly established that the RW model is the toughest benchmark in exchange rate forecasting. Almost all the studies on forecasting exchange rates use the RW benchmark (see, e.g. Corte *et al.* 2009; Rossi 2013; Della Corte *et al.* 2016; Filippou *et al.* 2023; Han *et al.* 2024). That is, previous studies show that the RW model is preferred for forecasting exchange rates. This model posits that there is no change in exchange rates between month t and $t + 1$,

$$S_{i,t+1} = S_{i,t} + \varepsilon_{i,t+1} \quad (7)$$

The out-of-sample returns based on the RW model can be calculated as:

$$\hat{r}_{i,t+1} = \log(\hat{S}_{i,t+1}) - \log(S_{i,t}) = \log(S_{i,t}) - \log(S_{i,t}) = 0. \quad (8)$$

But some studies also employ the random walk with drift (RWWD, hereafter) as the benchmark (see, e.g. Corte *et al.* 2009; Corte *et al.* 2016; Della Corte *et al.* 2016). The calculation involves regressing historical exchange rate changes on a constant or equal to the historical average change function, which implies constant changes in exchange rates. Specifically, it can be expressed as below.

$$S_{i,t+1} = S_{i,t} + c_i + \varepsilon_{i,t+1}, \quad (9)$$

where c_i denotes the constant change in exchange rates from month t to $t + 1$. Similarly, the out-of-sample return based on

the RWWD model is:

$$\begin{aligned} \hat{r}_{i,t+1} &= \log(\hat{S}_{i,t+1}) - \log(S_{i,t}) \\ &= \log(S_{i,t} + c_i) - \log(S_{i,t}) = c_i. \end{aligned} \quad (10)$$

In our study, we use both models as benchmark models and use the historical average changes as the RWWD model.

2.4. Forecasting evaluation

Following Campbell and Thompson (2008), we first use the out-of-sample R^2 statistic (R_{OS}^2) to evaluate the out-of-sample forecasting performance. Specifically, the R_{OS}^2 statistic can be expressed as:

$$R_{i,OS}^2 = 1 - \frac{\sum_{k=1}^{T-m} (r_{i,m+k} - \hat{r}_{i,m+k})^2}{\sum_{k=1}^{T-m} (r_{i,m+k} - \bar{r}_{i,m+k})^2}, \quad (11)$$

where $r_{i,m+k}$, $\bar{r}_{i,m+k}$, and $\hat{r}_{i,m+k}$ denote the actual returns, benchmark returns, and the predictive returns of the i -th currency for month $m + k$, respectively. And m stands for the number of in-sample observations. The R_{OS}^2 statistic shows how much the forecasting model's mean squared forecast error (MSFE) has decreased in comparison to the benchmark model, with a positive value indicating that the predictive model's performance surpasses that of the benchmark model.

The second forecast evaluation criterion is the MSFE-adjusted statistic proposed by Clark and West (2007). It tests the null hypothesis that the MSFE of the benchmark model is less than or equal to that of the predictive model, namely $R_{OS}^2 \leq 0$. The alternative hypothesis is that the MSFE of the predictive model is less than that of the benchmark model, namely $R_{OS}^2 > 0$. Specifically, it can be computed as follows:

$$f_{i,t} = (r_{i,t} - \bar{r}_{i,t})^2 - (r_{i,t} - \hat{r}_{i,t})^2 + (\bar{r}_{i,t} - \hat{r}_{i,t})^2, \quad (12)$$

where $r_{i,t}$, $\bar{r}_{i,t}$, and $\hat{r}_{i,t}$ denote the actual returns, benchmark model returns, and predictive model returns of the i -th currency, respectively. The Clark and West (2007) statistic, or the t -statistic, can be obtained by regressing $\{f_s\}_{s=m+1}^T$ on a constant.

3. Data

3.1. Oil price data

The price of West Texas Intermediate (WTI) is a benchmark price for other types of crude oil, and its price is influenced to some extent by the global supply and demand for crude oil. We collect daily closing price data for WTI oil from the U.S. Energy Information Administration.[†] The daily price data spans from January 02, 1985, to December 30, 2022. We then calculate monthly realized skewness of oil price returns.

[†] <https://www.eia.gov/dnav/pet/hist/RCLC1D.htm>

Table 1. Summary statistics.

	Min	Max	Mean	Median	Std	Skew	Kurt	CV
AUD	0.928	1.997	1.360	1.346	0.212	0.550	3.893	0.156
CAD	0.956	1.599	1.270	1.288	0.155	−0.105	2.366	0.122
CHF	0.780	2.808	1.278	1.252	0.330	1.344	6.239	0.258
DKK	4.734	11.803	6.536	6.389	1.063	1.789	8.224	0.163
GBP	0.483	0.915	0.645	0.634	0.082	0.523	3.089	0.127
JPY	76.655	260.340	118.050	112.921	27.228	2.503	12.549	0.231
NOK	5.058	10.578	7.201	6.935	1.166	0.508	2.417	0.162
NZD	1.151	2.508	1.614	1.548	0.286	0.908	3.564	0.177
SEK	5.286	11.129	7.672	7.583	1.239	0.471	2.558	0.161

The table displays the summary statistics of nine exchange rates. The entire period spans from January 1985 to December 2022. Exchange rates are denoted as the quantity of domestic currency units per US dollar. The last column is the coefficient of variation, defined as the ratio of standard deviation to mean.

3.2. Exchange rate data

Following Ren *et al.* (2021) and Liu *et al.* (2020), we use monthly average spot exchange rates for nine currencies: the Australian dollar (AUD), Canadian dollar (CAD), Swiss franc (CHF), Danish krone (DKK), British pound (GBP), Japanese yen (JPY), Norwegian krone (NOK), New Zealand dollar (NZD), and Swedish krona (SEK). The reason for employing the mean value is the significant volatility of exchange rates. Adopting a single trading day's value as the monthly representative (e.g. the first or last day of a month) could lead to substantial distortions due to unexpected market shocks. All of the data are obtained from the Organization for Economic Cooperation and Development (OCED) *Monthly Financial Statistics*[†] and the sample ranges from January 1985 to December 2022.

Table 1 reports the summary statistics for the exchange rate data. While the standard deviations of most currencies are relatively similar, the JPY exhibits a notably higher standard deviation of 27.228. Despite this elevated variability, the JPY also has a substantially larger mean value of 118.050. Consequently, when considering relative variability, the coefficient of variation (CV) for the JPY is calculated as 0.231, which is comparable to other currencies. Except for CAD, the exchange rates are right-skewed. It is well known that the kurtosis of the normal distribution is 3. Except for CAD, NOK, and SEK, the kurtosis of exchange rates in other countries is greater than 3, suggesting that these exchange rate distributions are characterized by leptokurtic, an important feature of financial markets (David 1997).

4. Empirical results

4.1. In-sample results

Table 2 reports the in-sample results with robust standard errors. All exchange rate returns (excluding JPY) show statistically significant negative correlations with realized skewness at the 5% level, with slope coefficients ranging from −0.271% to −0.530%. The limited predictive performance for the JPY can be attributed to two primary factors. First, the

Table 2. In-sample results.

	$\alpha_i(\%)$	t_α	$\beta_i(\%)$	t_β	$R^2(\%)$
AUD	0.022	0.146	−0.365***	−3.343	1.530
CAD	−0.009	−0.097	−0.271***	−3.644	2.322
CHF	−0.246**	−1.747	−0.293**	−2.221	1.021
DKK	−0.124	−0.878	−0.362***	−3.071	1.836
GBP	−0.042	−0.308	−0.468***	−4.538	3.164
JPY	−0.150	−0.945	−0.193	−1.272	0.441
NOK	−0.012	−0.087	−0.530***	−3.985	3.483
NZD	−0.086	−0.546	−0.365***	−3.202	1.480
SEK	0.012	0.078	−0.327***	−2.884	1.339

This table displays the in-sample results for $r_{i,t+1} = \alpha_i + \beta_i RSK_t + \varepsilon_{i,t+1}$, where $r_{i,t+1}$ denotes exchange rate returns, RSK_t denotes realized skewness of oil price returns, and $\varepsilon_{i,t+1}$ denotes the disturbance term. The table presents coefficient estimates alongside Newey and West (1987) t -statistics and R^2 , with coefficients and R^2 reported as percentages. The choice of bandwidth is based on a rule of thumb based on sample size. The detail is as follows: $m = \text{floor}(4 * (T/100) \wedge (2/9))$, where m is the bandwidth and T is the number of the whole sample. Asterisks (***, **, *) mark significance at the 1%, 5%, and 10% thresholds, respectively. The whole period spans from January 1985 to December 2022.

Bank of Japan's frequent interventions in currency markets exert a significant influence on yen exchange rate dynamics, often distorting predictable patterns. Second, the yen's role as a safe-haven currency adds further complexity, as it drives unpredictable capital flows, particularly during periods of heightened market turbulence. Consequently, forecasting the JPY's returns poses a considerable challenge. Theoretically, a negative or relatively lower skewness implies a higher downside probability of returns. Our results suggest that when realized skewness reflects a higher risk of decline for oil prices, the exchange rate returns tend to increase, which implies the appreciation of the USD. The results support the prior research on the negative correlation between oil price and USD (Reboredo *et al.* 2014; Liao *et al.* 2018).

In addition, the in-sample R^2 s except for JPY, are larger than the 0.5% benchmark proposed by Campbell and Thompson (2008), with the highest value reaching 3.483% for NOK. The evidence above confirms that realized skewness can effectively negatively forecast exchange rate returns.

[†] <https://stats.oecd.org/?lang=en>

Table 3. Out-of-sample results.

	Panel A: RWWD			Panel B: RW		
	R_{OS}^2 (%)	CW-stat	p-value	R_{OS}^2 (%)	CW-stat	p-value
AUD	1.737***	2.762	0.003	1.279**	2.209	0.014
CAD	2.436***	3.459	0.000	2.045***	2.893	0.002
CHF	0.350*	1.415	0.078	0.640**	2.003	0.023
DKK	1.399**	2.197	0.014	0.649**	1.928	0.027
GBP	2.966***	3.456	0.000	2.094***	3.020	0.001
JPY	-1.076	0.290	0.386	-4.094	-0.142	0.557
NOK	3.386***	3.026	0.001	3.018***	2.905	0.002
NZD	1.130**	2.001	0.023	0.849**	1.718	0.043
SEK	1.102**	2.044	0.020	0.764**	1.789	0.037

This table reports the out-of-sample performance across various exchange rates by expanding windows, presenting the out-of-sample R^2 (R_{OS}^2), the Clark and West (2007) MSFE-adjusted statistic, and its p -value. The null hypothesis of Clark and West (2007) test is that the MSFE of the benchmark model does not exceed that of the forecast model, while the alternative suggests the opposite. Asterisks (***, **, *) mark significance at the 1%, 5%, and 10% thresholds, respectively. The in-sample period spans January 1985 to December 1999, with the out-of-sample period extending from January 2000 to December 2022.

4.2. Out-of-sample results

Table 3 reports the out-of-sample performance using the initial 15 years as the estimation period and generates the exchange rate forecasts by the expanding window. In line with the in-sample findings, the realized skewness-based forecasts outperform the random walk benchmark model. When employing either RWWD or RW as the benchmark, all exchange rate returns except for JPY display significant positive R_{OS}^2 s at least at the 10% level. Specifically, realized skewness provides the highest R_{OS}^2 , reaching 3.386% (3.018%) for NOK when using the RWWD (RW) as the benchmark. The out-of-sample results further confirm the predictive ability of realized skewness for exchange rate returns.

4.3. Cumulative squared forecast error

To provide a more intuitive picture of the out-of-sample predictive power of the model at each moment, we provide a plot of the cumulative squared forecast error (CSFE) of the predictive models with respect to the RW model in Figure 1.† The CSFE is calculated as,

$$CSFE_i = \sum_{t=m+1}^q (e_{bench_{i,t}}^2 - e_{mod_{el_{i,t}}}^2) \quad (13)$$

where $e_{bench_{i,t}}$ and $e_{mod_{el_{i,t}}}$ denote the forecast error of the benchmark model and the realized skewness-based predictive model of the i -th currency at time t , respectively. We choose the RW model as the benchmark model. If the curve of CSFE increases from time t to $t+1$, the predictive performance of the corresponding predictive model surpasses that of the benchmark model at time $t+1$.

According to Figure 1, we can see that the curves of the forecasting models for eight of the nine currencies (except for JPY) show a general upward trend, despite a decline in some periods. This suggests that the MSFE of the forecasting model considering realized skewness is smaller than that of the benchmark model. Hence, the forecasting model with

realized skewness demonstrates superior predictive performance compared to the benchmark model across the majority of out-of-sample periods. In summary, incorporating realized skewness significantly enhances exchange rate return predictability.

4.4. Economic significance

It should be acknowledged that although the results may be statistically notable, they may not hold economic significance. To further evaluate the economic value of our forecasting results, we build a simplified trading strategy as suggested by Iyke *et al.* (2022) and Ren *et al.* (2021). This strategy focuses on a single currency and involves buying or selling one unit of the foreign currency based on the forecasting appreciation or depreciation of the currency by the benchmark or predictive models. To ensure that our analysis is meaningful, we only consider the RWWD as the benchmark model, as the RW model does not offer any effective information regarding changes in the exchange rate.

Specifically, we first compute the original returns of the trading strategies, which can be expressed as:

$$r_{i,t} = \begin{cases} r_{i,t}, & \text{if } \hat{r}_{i,t} \geq 0 \\ -r_{i,t}, & \text{otherwise} \end{cases} \quad (14)$$

We calculate the mean of the original returns and multiply this value by 1200 to obtain the annualized average returns in percentage. We use the t -test to test whether original returns are significantly greater than 0. Second, we calculate the certainty equivalent return (CER) for different trading strategies. The CER is computed as follows:

$$CER_i = \hat{\mu}_{i,P} - 0.5\gamma\hat{\sigma}_{i,P}^2, \quad (15)$$

where $\hat{\mu}_{i,P}$ and $\hat{\sigma}_{i,P}$ denote the mean and variance of the trading strategy during the out-of-sample period, respectively. And γ is the risk aversion coefficient of investors, which is equal to 3. CER gain denotes the CER difference between the predictive model with realized skewness and RWWD. We multiply this difference by 1200 to obtain the annualized CER gain in percentage.

† When using the RWWD as the benchmark model, the results are similar.

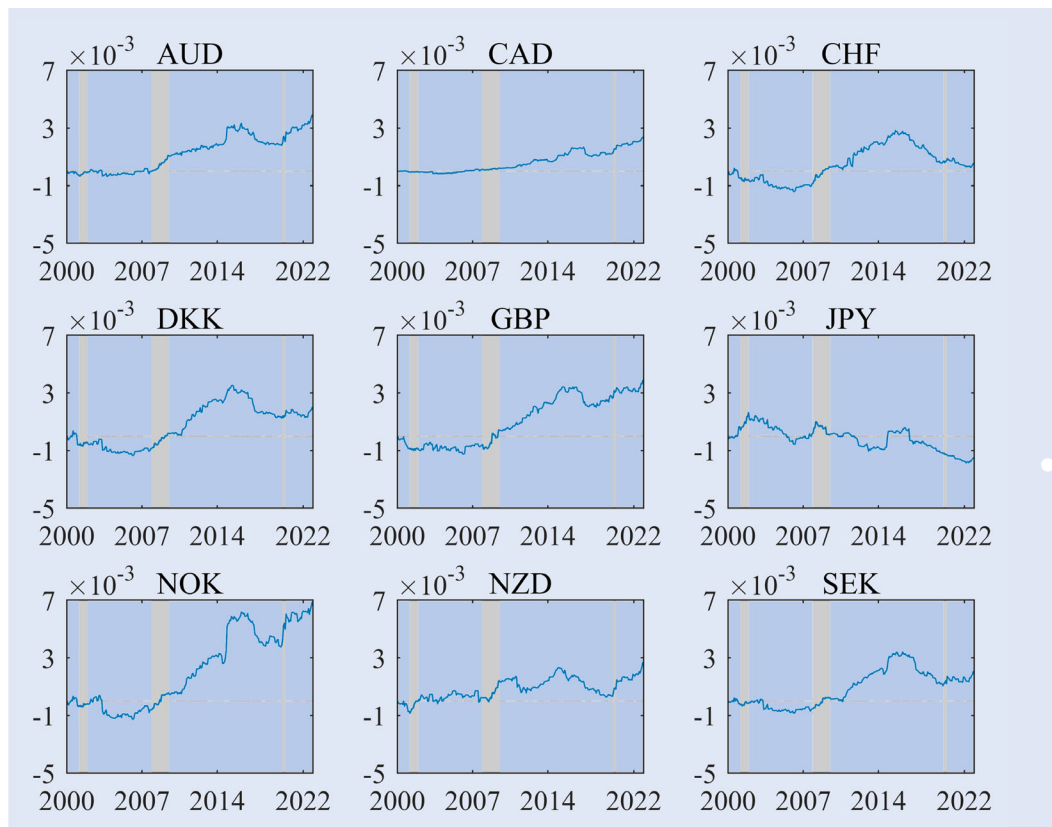


Figure 1. Cumulative square forecast error, The figure plots the cumulative square forecast error (CSFE) of forecasting models compared to the benchmark model (RW). A rising curve indicates that the predictive accuracy of the respective forecasting model surpasses that of the RW model. Vertical bars depict recessions as dated by the National Bureau of Economic Research (NBER). The sample period covers January 2000 to December 2022.

Third, we compute the Sharpe ratio of the trading strategies, which is given by.

$$SR_i = \frac{\bar{r}_{i,p}^e}{\sigma_{i,p}^e}, \quad (16)$$

where $\bar{r}_{i,p}^e$ and $\sigma_{i,p}^e$ are the mean and standard deviation of the trading strategies' excess return over the out-of-sample period. We obtain the SR gain, which is the SR difference between the predictive model considering realized skewness and RWWD. Similarly, we multiply the difference by $\sqrt{12}$ to obtain the annualized difference. Additionally, we also compute the statistical significance of the CER gain and SR gain according to DeMiguel *et al.* (2009), Jobson and Korkie (1981), and Memmel (2003).

Fourth, we consider the maximum drawdown, which is an indicator of downside risk over a given period. It calculates the maximum cumulative loss that can be incurred between the investing strategies' peak and the subsequent trough. A relatively low value of the maximum drawdown is a successful sign for investment strategies since it means smaller losses from the investment.

Table 4 presents the annualized average return and maximum drawdowns of the trading strategies together with their matching CER gains and SR gains. In Panel A, except for CHF and JPY, the remaining seven currencies all generate positive average returns that are far superior to those of the benchmark models in Panel B, with the maximum value reaching 3.554% for AUD. Compared to Panel B, Panel A

demonstrates a substantial reduction in the maximum drawdown for the same set of seven currencies. Moreover, four of the currencies generate returns that are significantly greater than zero under the strategy employing realized skewness in the predictive model. In Panel B, only the CHF, DKK, and NZD generate positive returns, with the CHF generating the largest values of 2.302%. In Panel C, 7 out of 9 currencies (except for CHF and JPY) exhibit positive CER gains and SR gains. This suggests that the predictive model considering realized skewness can yield higher economic gains than the benchmark model in most currencies. The AUD achieves the maximum CER gain of 5.738% and the highest Sharpe ratio of 0.588. Notably, 4 out of them demonstrate statistical significance.

In summary, for most currencies, the predictive model based on realized skewness can provide higher annualized average returns, CER gains, SR gains, and lower maximum drawdowns. These results provide further confirmation of the predictability of realized skewness on exchange rate gains in an economic sense.

In addition, we provide the cumulative returns generated by the trading strategy under the realized skewness-based predictive model and the RWWD model, respectively.[†] The results are shown in Figure 2. The blue (red) line represents the cumulative returns using realized skewness (the RWWD) for currency return forecasts given in one of nine currencies.

[†] We thank the anonymous referee for this valuable suggestion.

Table 4. Investment strategy for individual currencies.

	Panel A: RSk		Panel B: RWWD		Panel C: Economic gain	
	Avg.	MDD	Avg.	MDD	CER gain	SR gain
AUD	3.554** (1.745)	27.289	− 2.163 (− 1.059)	92.470	5.728*** (2.461)	0.588*** (2.456)
CAD	2.165* (1.605)	19.886	− 1.583 (− 1.171)	59.202	3.751*** (2.884)	0.579*** (2.872)
CHF	0.702 (0.43)	28.864	2.302* (1.415)	26.737	− 1.607 (− 1.555)	− 0.205 (− 1.553)
DKK	1.960 (1.223)	19.972	0.200 (0.125)	46.972	1.764 (1.103)	0.229 (1.102)
GBP	2.493* (1.619)	20.723	− 1.227 (− 0.794)	60.554	3.726** (2.009)	0.503** (2.005)
JPY	− 1.406 (− 0.861)	58.009	− 1.186 (− 0.726)	65.140	− 0.219 (− 0.182)	− 0.028 (− 0.183)
NOK	2.765* (1.426)	26.812	− 1.722 (− 0.886)	75.985	4.494** (2.009)	0.482** (2.007)
NZD	2.579 (1.241)	39.319	0.468 (0.225)	43.881	2.119 (0.905)	0.212 (0.905)
SEK	1.450 (0.783)	35.098	− 0.591 (− 0.319)	50.167	2.043 (0.912)	0.230 (0.912)

This table reports the performance of the trading strategy that initiates long (short) positions in foreign currency when the benchmark models or predictive models forecast an appreciation (depreciation) of the foreign currency. Panels A and B are the results of the predictive model with realized skewness and the benchmark model (i.e. the random walk with drift, RWWD), respectively. 'Avg.' denotes the annualized average returns, which are in percentage. The brackets below estimate report *t*-statistics for the returns generated by the trading strategies, which assess whether the returns are statistically greater than zero. 'MDD' is the maximum cumulative loss from the peak to the subsequent trough of the strategy based on the RWWD or predictive models. Panel C denotes the results of economic gains. 'CER gain' and 'SR gain' denote the annualized certainty equivalent return (CER) difference and the annualized Sharpe ratio difference between the predictive model with realized skewness and the RWWD, respectively. The brackets below report statistics to test the statistical significance of the CER gain and SR gain. Asterisks (***, **, *) mark significance at the 1%, 5%, and 10% thresholds, respectively.

There are two main findings. First, in most currencies (except for JPY), the cumulative returns of trading strategies based on our predictive model show a more consistent growth trend, while the benchmark models show a more volatile trend. In the JPY and CHF, our predictive model underperforms the benchmark model, which is consistent with the results of economic tests. Second, during recessions, our forecasting models typically show more resilient and stable performance, while the benchmark models show greater volatility. As a result, the cumulative returns of the predictive model with realized skewness have a stronger long-term upward trend compared to that of the benchmark model.

5. Economic explanation

5.1. Inflation rates, investor sentiment, and risk transmission channels

We consider three channels—(a) inflation rates, (b) investor sentiment, and (c) risk transmission—that explain the predictive power of crude oil's realized skewness on subsequent exchange rate returns.

First, the realized skewness of oil returns may drive the exchange rate movements by influencing inflation rates among countries. As one of the key commodities, the increase in oil prices historically precedes high inflation and low economic growth (see, e.g. Barsky and Kilian 2004; Chen 2009; Choi *et al.* 2018; Kilian and Zhou 2022). Specifically, realized skewness reflects the risk of upward or downward in oil prices, and oil price fluctuations can affect the price level through both its energy and non-energy components (Conflitti

and Luciani 2019). In addition, changes in inflation rates may vary across countries depending on their share of the CPI basket, energy subsidies, dependence on oil, and production costs in exporting countries (Sek *et al.* 2015; Choi *et al.* 2018). According to the purchasing power parity theory, inflation rate disparities between two nations predict currency depreciation for the higher-inflation country and appreciation for the lower-inflation one.

Second, realized skewness of oil price returns may reflect investor sentiment and trading preferences for the exchange rate. Behavioral finance theory argues that investors' investments are influenced by their psychological and emotional factors. Research shows investors favor assets with positive skewness, akin to lottery assets (see, e.g. Kraus and Litzenberger 1976; Harvey and Siddique 2000; Clifton Green and Hwang 2012). Consequently, such assets are often overpriced (see, e.g. Mitton and Vorkink 2007; Bali *et al.* 2011; Conrad *et al.* 2014), particularly amid high investor sentiment (Stambaugh *et al.* 2012). This behavioral bias can impact exchange rate movements.

Third, realized skewness of oil price returns may contain information on global risks that affect the exchange rate. Tail risk—especially negative skewness—does not only manifest in individual assets but may also reflect the contagion effects of systemic risk. Existing studies (Kelly and Jiang 2014; Neuberger and Payne 2021) have also shown that tail risk can serve as an important indicator of systemic market risk, with skewness directly quantifying the characteristics of tail risk. Therefore, realized skewness can be a powerful signal of global risk. Sarno *et al.* (2012) state that the foreign exchange risk premium and global risk are closely connected. During periods of global market uncertainty, investors typically seek

higher risk premiums for high-yield currencies and lower premiums for low-yield ones. Brunnermeier *et al.* (2008) argue that the CBOE Volatility Index (VIX) can be viewed as a useful indicator of global risk appetite, with a higher VIX predicting higher returns in high-rate currencies and lower returns in low-rate currencies. In other words, the increase in global risk appetite will lead investors to demand different risk premiums for currencies with different yields, leading to different changes in the yields of the corresponding currencies, which finally cause changes in the exchange rate.

We employ the Consumer Price Index (CPI) as the proxy for the inflation rate, the University of Michigan Consumer Confidence Index (MSCI) as the proxy for investor sentiment,[†] and the VIX as the proxy for aggregate risk.[‡]

Specifically, since the exchange rate is denominated relative to the dollar, for CPI, we first compute the growth rates by taking the first logarithmic differences and use the differentials between other countries and the United States ($\Delta CPI_t - \Delta CPI_t^*$) as economic indicators.

$$s_proxy_{i,t} = \Delta CPI_t - \Delta CPI_t^*, \quad (17)$$

ΔCPI_t^* denotes the inflation rates of the United States and ΔCPI_t denotes the inflation rates of other countries. For MSCI and VIX, we multiply the raw values by the signs of the differential as indicators (Sarno *et al.* 2012; Han *et al.* 2024),

$$s_proxy_{i,t} = proxy_t * sign[r_{i,t} - r_t^*], \quad (18)$$

Where $proxy_t$ denotes the raw values of MSCI or VIX in month t , $r_{i,t}$ denotes the exchange rate returns of i -th currencies, and r_t^* denotes the domestic (US dollar) exchange rate returns. The predictive regression can be specified as:

$$s_proxy_{i,t+1} = \alpha_i + \beta_i RSk_t + \varepsilon_{i,t+1}. \quad (19)$$

Table 5 reports the coefficients and corresponding t -statistics estimated from the predictive regressions of realized skewness on individual indicators.

In Panel A, realized skewness of oil price returns in 6 of 9 currencies generates a significantly negative correlation with the differential in inflation rates,[§] which indicates that

[†] However, we don't find a significant impact of skewness on Baker and Wurgler's sentiment (BW) index, which is also widely used to measure investor sentiment. A potential explanation for this failure may be that the MSCI mainly measures individual investor sentiment while the BW index reflects market-wide investor sentiment (Baker and Wurgler 2006; Schmeling 2009; Fong 2015), and only the extreme negative values of skewness can exert influence on the BW index (Zhang *et al.* 2021).

[‡] The data for CPI is obtained from OCED, which is available at the website <https://data.oecd.org/price/inflation-cpi.htm>. The data for MSCI and VIX are obtained from FRED, which is available at the websites <https://fred.stlouisfed.org/series/UMCSENT> and <https://fred.stlouisfed.org/series/VIXCLS>.

[§] For Australia and New Zealand, where the CPI data is reported quarterly, we interpolate monthly observations using linear interpolation. Therefore, the monthly CPI data we construct are noisy causing the corresponding results to be insignificant. The significant prediction of skewness on Japan's CPI do not contradict the unpredictability of skewness on the JPY. The empirical results show that when using Japan's CPI to forecast the yen's exchange rate, the

realized skewness may be a strong indicator for future country economic expectations. When realized skewness falls or becomes negative, the differential in the inflation rate will rise. According to the definition of the differential in inflation rate, higher inflation rates in other countries lead to currency depreciation, while lower inflation rates in the United States lead to appreciation of the dollar, which increases exchange rate returns.

In Panels B and C, the realized skewness of oil price returns exhibits uniformly significant negative correlations with MSCI and VIX, except for JPY and CHF, which is consistent with their out-of-sample performance in Section 4.2. In other words, as an early signal of future trends in the oil market, the decrease in realized skewness will significantly raise investors' sentiment and global risk appetite, thus exerting predictability in exchange rate returns.

Taken together, realized skewness of oil price returns can generate predictability in exchange rate returns by influencing inflation rates, investor sentiment, and global risks.

5.2. The information content of realized skewness

In the standard high-frequency econometric framework, realized skewness asymptotically depends only on jumps, but in finite samples, it is also influenced by drift (Bandi *et al.* 2023). To discern whether the observed predictability originates from jumps or the drift component, we incorporate measures of signed jumps and drift into our regression models at the high-frequency level.[¶] The specific extended model is as follows:

$$r_{i,t+1} = \alpha_i + \beta_i RSk_t + \beta_{i,+} Jump_t^+ + \beta_{i,-} Jump_t^- + \beta_{i,d} Drift_{i,t} + \varepsilon_{i,t+1}, \quad (20)$$

where $Jump_t^+$ and $Jump_t^-$ are the sum of the positive and negative jumps, respectively. Much of the literature provides information on the detection and estimation of jumps (see, e.g. Andersen *et al.* 2007; Corsi *et al.* 2010; Patton and Sheppard 2015; Kolokolov 2022). We follow Bandi *et al.* (2023) and the jump is based on the truncation method, which distinguishes continuous price changes from jumps by setting a threshold. The measure of the jump is defined as the monthly sum of intraday returns whose absolute values exceed the threshold. The threshold is calculated regarding Bollerslev and Todorov (2011) and Aleti *et al.* (2025). The drift component is the past monthly return (Amaya *et al.* 2015). In the high-frequency level, the monthly realized skewness is the sum of the daily realized skewness for all trading days in the month, i.e. $RSk_t = \frac{1}{N_t} \sum_{i=1}^{N_t} RDSk_{t-i}$, where $RDSk_i = \frac{\sqrt{M} \sum_{j=1}^M r_{ij}^3}{RDV_i^{1.5}}$ is the realized skewness of day i and $M = 1/\Delta$, Δ is the 5-min sampling frequency.^{||}

results are not significant, while other countries are significant. This suggests that the predictability of JPY cannot be explained through the inflation rate channel. The empirical results are available upon request.

[¶] We thank the associate editor for this valuable suggestion.

^{||} Due to the limitations of high-frequency data, the full sample is from January 2008 to March 2024.

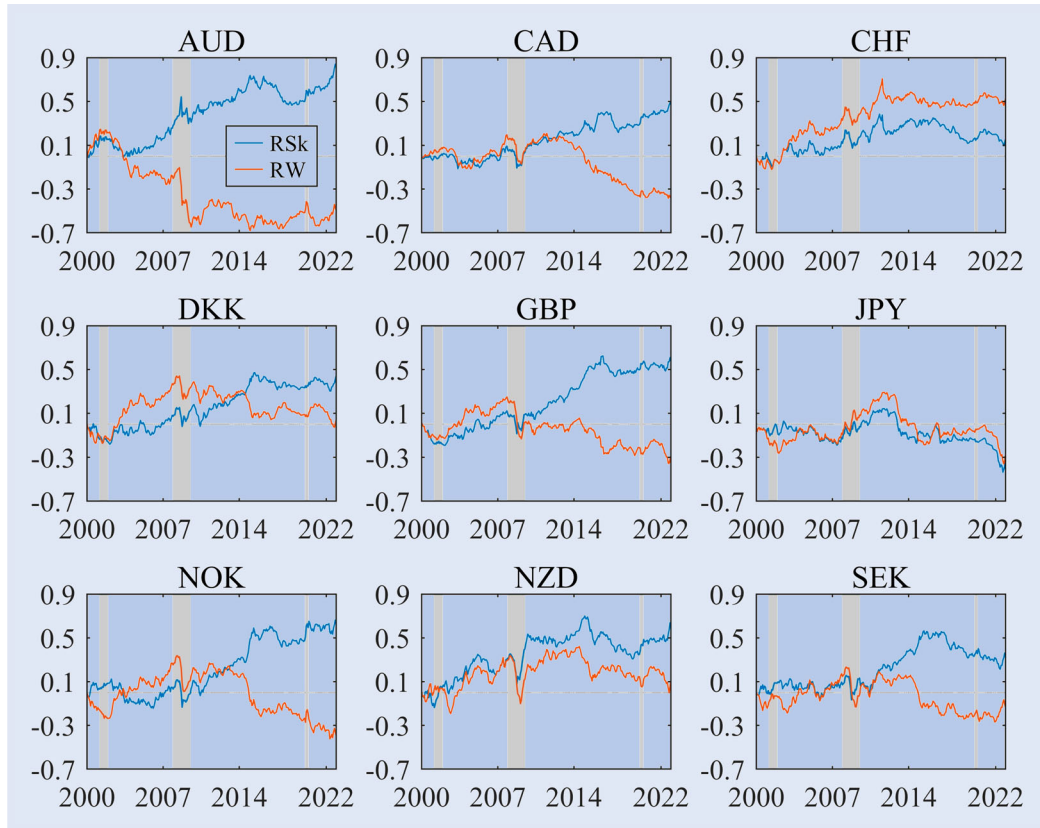


Figure 2. Cumulative returns generated by the trading strategy. The figure plots the cumulative return generated by the trading strategy under different predictions. The blue (red) line represents the cumulative returns using realized skewness (the random walk with drift) for currency return forecasts given in the panel heading. Vertical bars depict recessions as dated by the National Bureau of Economic Research (NBER). The sample period covers January 2000 to December 2022.

Table 5. The results of the economic explanation.

	Panel A: CPI		Panel B: MSCI		Panel C: VIX	
	Coef (%)	t-value	Coef (%)	t-value	Coef (%)	t-value
AUD	−0.069	−1.313	−14.243***	−3.339	−4.027***	−3.358
CAD	−0.046***	−2.662	−13.176***	−2.981	−3.625***	−3.145
CHF	−0.063***	−3.498	−1.624	−0.333	−1.566	−1.260
DKK	−0.031	−1.336	−9.390**	−1.969	−4.171***	−3.944
GBP	−0.073***	−2.916	−13.516***	−2.961	−4.355***	−3.909
JPY	−0.067***	−3.198	5.509	1.059	1.271	0.973
NOK	−0.085***	−3.463	−14.363***	−3.371	−4.741***	−4.468
NZD	−0.082	−1.576	−13.127***	−3.259	−3.740***	−3.505
SEK	−0.098***	−3.019	−8.251***	−1.753	−2.974***	−2.608

The table displays the results of predictive regression $s_proxy_{i,t+1} = \alpha_i + \beta_i RSk_t + \varepsilon_{i,t+1}$. In Panel A, $s_proxy_{i,t} = \Delta CPI_t - \Delta CPI_t^*$ denotes the differential of the inflation rate. In Panels B and C, $s_proxy_{i,t} = proxy_t * sign[r_{i,t} - r_t^*]$, $proxy_t$ denotes investor sentiment and VIX, respectively. We employ the Consumer Price Index (CPI) to represent inflation, the University of Michigan Consumer Confidence Index (MSCI) to gauge investor sentiment, and VIX to measure overall risk. The VIX data covers the period from January 1990 to December 2022, whereas the CPI and MSCI data extend from January 1985 to December 2022. For Australia and New Zealand, where the CPI data is reported quarterly, we interpolate monthly observations using linear interpolation. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 6 reports the in-sample results of Eq.(4) and Eq.(20) at the level of 5-min high-frequency returns. Panel A shows that realized skewness of crude oil return negatively predicts all exchange rate returns except for JPY, with five of them being significantly negative.[†] In Panel B, we add the sign

[†] Compared to Table 2, the regression coefficients for CHF, DKK, and GBP became less significant. The reason may be that the smaller sample size leads to a reduction in the efficacy of the statistical test, making the significance level decrease.

jumps and drift components to our regression model. The coefficients of realized skewness become rather insignificant or even positive. The drift terms have very significant negative coefficients (except for predictions involving CHF and JPY), representing short-term reversals. Both positive and negative jumps exhibit a pronounced negative sign (except for predictions involving CHF and JPY), which is economically meaningful. This negative sign can be interpreted similarly to the negative skewness: market participants are generally assumed to dread substantial negative returns while favoring significant

Table 6. The in-sample results of the sign jumps and drift component.

	Panel A: Univariate model		Panel B: Multivariate model			
	$\alpha_i(\%)$	$\beta_i(\%)$	$\beta_{i,RSk}(\%)$	$\beta_{i,+}(\%)$	$\beta_{i,-}(\%)$	$\beta_{i,d}(\%)$
AUD	0.198 (0.769)	-1.595*** (-3.108)	-0.356 (-0.691)	-5.758*** (-3.108)	-5.146*** (-2.996)	-7.902*** (-2.805)
CAD	0.179 (1.132)	-1.007*** (-2.538)	-0.168 (-0.440)	-3.783*** (-2.538)	-3.418*** (-2.969)	-6.005*** (-2.964)
CHF	-0.101 (-0.610)	-0.353 (-0.751)	-0.039 (-0.074)	-1.365 (-0.751)	-1.172 (-1.205)	-2.644 (-1.084)
DKK	0.166 (0.867)	-0.331 (-0.796)	0.447 (1.025)	-3.574*** (-0.796)	-3.400*** (-2.758)	-4.995* (-1.613)
GBP	0.233 (1.093)	-0.252 (-0.461)	0.769* (1.385)	-4.325** (-0.461)	-4.511** (-2.271)	-8.426*** (-2.580)
JPY	0.161 (0.742)	0.293 (0.525)	0.043 (0.077)	0.475 (0.525)	1.045 (1.059)	5.200* (1.582)
NOK	0.393** (1.660)	-1.619*** (-2.849)	-0.271 (-0.512)	-6.555*** (-2.849)	-6.171*** (-4.139)	-6.434** (-1.734)
NZD	0.144 (0.566)	-0.735* (-1.334)	0.305 (0.509)	-4.152*** (-1.334)	-3.799*** (-2.380)	-10.787*** (-3.566)
SEK	0.273 (1.134)	-0.803* (-1.586)	0.271 (0.484)	-4.855*** (-1.586)	-4.620*** (-3.183)	-7.368*** (-2.486)

This table displays the in-sample results at the high-frequency level. Panels A and B are the results of $r_{i,t+1} = \alpha_i + \beta_i RSK_t + \varepsilon_{i,t+1}$ and $r_{i,t+1} = \alpha_i + \beta_{i,RSk} RSK_t + \beta_{i,+} Jump_t^+ + \beta_{i,-} Jump_t^- + \beta_{i,d} Drift_{i,t} + \varepsilon_{i,t+1}$, respectively. $r_{i,t+1}$ denotes exchange rate returns, RSK_t denotes the monthly realized skewness of oil price returns, which is the sum of the daily realized skewness for all trading days in the month. $Jump_t^+$ and $Jump_t^-$ are the sum of the positive and negative jumps, respectively. The measure of the jump is defined as the monthly sum of intraday returns whose absolute values exceed the threshold. $Drift_{i,t}$ is the drift component, which is the past monthly return. $\varepsilon_{i,t+1}$ denotes the disturbance term. The table presents coefficient estimates, with coefficients reported as percentages. The brackets below estimate report Newey and West (1987) t -statistics. The choice of bandwidth is based on a rule of thumb based on sample size. The detail is as follows: $m = \text{floor}(4 * (T/100) \wedge (2/9))$, where m is the bandwidth and T is the number of the whole sample. Asterisks (***, **, *) mark significance at the 1%, 5%, and 10% thresholds, respectively. Due to the limitations of high-frequency data, the whole period spans from January 2008 to March 2024.

positive returns. This suggests that the predictability of realized skewness for exchange rates stems from the jump and the drift components, which coincides with the findings of Bandi *et al.* (2023) in the cross-section.

6. Robustness and extensions

6.1. Other currencies

To ensure that the predictability of realized skewness is robust, we further select a range of currencies from developing countries and repeat the prediction procedure. Specifically, we choose the currencies of Brazil (BRL), China (CNY), India (INR), Russia (RUB), and South Africa (ZAR), which are known as the BRICS, as the sample for the robustness test. The whole sample period spans from January 1994 to December 2022, and we set the first 10 years as the estimation period. The exchange rate data is obtained from OCED *Monthly Financial Statistics*.[†]

Table 7 reports the out-of-sample results for the BRICS currencies. Apart from the BRL, the R_{OS}^2 of remaining four BRICS currencies are significantly positive, where the maximum value generated by the RWD (RW) as the benchmark model is 6.153% (4.416%) for the RUB (ZAR). Thus, we conclude that currency selection minimally impacts the forecasting power of realized skewness on exchange rate returns.

6.2. Different window sizes

Given that the sample split point may have an impact on the out-of-sample results, we choose alternative window sizes of 60, 120, and 240 observations as estimation periods and generate the out-of-sample forecasts by the expanding window again. For brevity, we only report the results based on the RW benchmark model.

Table 8 shows out-of-sample results varying by initial estimation window size, revealing that performance declines as the window shortens. However, when using the shortest window size of 60 observations, 5 of 9 currencies still generate significant positive R_{OS}^2 s. According to Chen *et al.* (2010) and Chen and Rogoff (2003), most of them belong to 'commodity currencies', with the highest R_{OS}^2 reaching 2.261% for NOK. Moreover, the R_{OS}^2 s become positive and more significant as the estimation window increases. This is the evidence that our model outperforms the benchmark model in different window sizes.

6.3. Rolling window

Clark and McCracken (2009) argue that financial time series usually have structural breaks, and thus the rolling windows are used with more frequency than the expanding window. Thereby, we further employ the rolling estimation windows to explore the predictability of realized skewness. Specifically, the rolling estimation windows generate out-of-sample forecasts by adding a new observation from the latest period and abandoning a previous observation from the most distant period. We choose the rolling estimation window sizes of 120, 180, and 240 observations and report

[†] <https://stats.oecd.org/?lang=en>

Table 7. The out-of-sample results of other currencies.

	Panel A: RWWD			Panel B: RW		
	$R_{OS}^2(\%)$	CW-stat	p -value	$R_{OS}^2(\%)$	CW-stat	p -value
BRL	3.940***	3.524	0.000	− 7.035	1.497	0.067
CNY	0.782*	1.608	0.054	1.097**	1.894	0.029
INR	0.667*	1.404	0.080	2.252***	2.398	0.008
RUB	6.153***	3.424	0.000	0.929***	2.524	0.006
ZAR	3.708***	3.619	0.000	4.416***	3.180	0.001

The table displays the out-of-sample findings for developing countries by expanding windows. Specifically, we focus on the currencies of Brazil, China, India, Russia, and South Africa, which are known as the BRICS. ***, **, and * indicate significance at the 1%, 5%, and 10% thresholds, respectively. The whole sample period covers January 1994 to December 2022, with the in-sample period spanning from January 1994 to December 2003.

Table 8. The out-of-sample results for different window sizes.

	Panel A: $m = 60$		Panel B: $m = 120$		Panel C: $m = 240$	
	$R_{OS}^2(\%)$	CW-stat	$R_{OS}^2(\%)$	CW-stat	$R_{OS}^2(\%)$	CW-stat
AUD	1.094**	2.302	1.271***	2.355	1.349***	2.404
CAD	1.449***	2.639	1.821***	2.834	1.831***	2.789
CHF	− 0.827	1.726	− 1.788	1.029	− 0.457	1.601
DKK	− 0.717	1.831	− 1.156	1.323	− 0.270	1.621
GBP	1.449***	2.688	1.190***	3.148	1.950***	3.306
JPY	− 2.477	1.068	− 4.715	− 0.244	− 3.781	− 0.046
NOK	2.261***	3.065	2.189***	2.757	2.875***	3.020
NZD	0.625**	1.994	0.451*	1.604	0.434*	1.487
SEK	− 0.331	1.269	0.108*	1.472	0.477**	1.685

The table displays the out-of-sample R^2 (R_{OS}^2), its Clark and West (2007) MSFE-adjusted statistic, and the corresponding p -value based on the different window sizes. Panels A, B, and C select 60, 120, and 240 observations, respectively, as alternative window sizes for the estimation period and generate out-of-sample results by an expanding window. For brevity, we only report the results based on the RW benchmark model. ***, **, and * indicate significance at the 1%, 5%, and 10% thresholds, respectively. The whole sample period spans from January 1985 to December 2022.

the out-of-sample results in Table 9. For brevity, we only report the out-of-sample results based on the RW benchmark model.

Consistently, the out-of-sample results become inferior when using the shorter rolling estimation windows. When using the rolling window of 120 observations, 4 of 9 currencies, including most commodity currencies, generate significant positive R_{OS}^2 s, and the NOK presents the maximum of 1.681%. Similarly, the R_{OS}^2 s become positive and significant as the window size expands, except for JPY.

Table 10 reports the results of annualized average returns based on various rolling window sizes. We choose the RWWD

as the benchmark model. As the window size increases, the average annualized returns generated by trading strategies using the predictive model with realized skewness increase in most currencies. When the rolling window size is 240 observations, 6 of 9 currencies generate returns that are significantly greater than zero, with the maximum value reaching 6.353% for NOK. Under the benchmark model's trading strategy, while a limited number of currencies generated positive annualized returns, none achieved statistical significance at conventional levels.

In summary, our results suggest that predictability is robust for almost half of the nine currencies, especially for

Table 9. The out-of-sample results based on a rolling window.

	Panel A: $m = 120$		Panel B: $m = 180$		Panel C: $m = 240$	
	$R_{OS}^2(\%)$	CW-stat	$R_{OS}^2(\%)$	CW-stat	$R_{OS}^2(\%)$	CW-stat
AUD	0.352**	1.909	0.584***	2.603	1.974***	2.462
CAD	1.042***	2.375	1.410***	3.229	3.205***	3.187
CHF	− 2.421	1.107	− 2.712	0.673	1.327**	2.003
DKK	− 2.371	1.562	− 1.044	1.983	2.733***	2.756
GBP	0.287***	3.455	0.911***	3.383	4.403***	3.57
JPY	− 2.429	0.095	− 2.354	0.431	− 2.132	− 0.233
NOK	1.681***	2.574	1.934***	2.876	4.862***	3.317
NZD	− 0.040	1.418	− 0.371	1.533	0.552*	1.363
SEK	− 1.030	1.633	− 1.146	1.461	1.362**	2.068

This table displays the out-of-sample performance using rolling windows with initial sizes of 120, 180, and 240. The rolling window always contains a specific number of samples and generates the out-of-sample results by adding new observations from the latest period and abandoning the first previous observations. For brevity, we only report the results based on the RW benchmark model. The R_{OS}^2 , MSFE-adjusted statistics and the p -value are shown in the table. ***, **, and * indicate significance at the 1%, 5%, and 10% thresholds, respectively. The whole sample period spans from January 1985 to December 2022.

Table 10. The economic test based on a rolling window.

	Panel A: $m = 120$		Panel B: $m = 180$		Panel C: $m = 240$	
	RSk	RWWD	RSk	RWWD	RSk	RWWD
AUD	1.866 (1.055)	− 0.770 (− 0.435)	2.658* (1.302)	− 2.892 (− 1.418)	4.153** (1.818)	− 2.350 (− 1.024)
CAD	1.915** (1.666)	0.209 (0.181)	2.013* (1.491)	− 1.824 (− 1.350)	2.272* (1.453)	− 1.666 (− 1.063)
CHF	0.412 (0.270)	0.852 (0.559)	0.657 (0.403)	− 0.048 (− 0.030)	4.827*** (2.745)	1.145 (0.641)
DKK	0.275 (0.189)	− 0.370 (− 0.255)	2.519* (1.575)	− 1.606 (− 1.002)	4.099*** (2.409)	− 1.224 (− 0.711)
GBP	3.216*** (2.427)	0.444 (0.332)	2.499* (1.623)	− 3.367 (− 2.195)	4.551*** (2.585)	0.452 (0.253)
JPY	− 0.118 (− 0.072)	− 0.190 (− 0.115)	− 0.808 (− 0.494)	− 0.599 (− 0.367)	0.035 (0.019)	− 0.225 (− 0.121)
NOK	3.176** (1.902)	0.446 (0.266)	3.737** (1.933)	− 3.639 (− 1.882)	6.353*** (2.910)	− 2.146 (− 0.966)
NZD	2.645* (1.479)	0.721 (0.402)	0.592 (0.284)	− 2.809 (− 1.352)	2.577 (1.130)	− 1.397 (− 0.611)
SEK	0.674 (0.417)	1.176 (0.728)	− 0.571 (− 0.308)	− 3.883 (− 2.112)	2.456 (1.182)	− 3.185 (− 1.536)

The table reports the annualized average returns in percentage and t -statistics for the returns generated by the trading strategies, which assess whether the returns are statistically greater than zero. The trading strategies involve taking a long (short) position in the foreign currency when the benchmark models (i.e. the random walk with drift) or predictive models forecast an appreciation (depreciation) of the foreign currency. The brackets below estimate report t -statistics. Panels A, B, and C represent rolling window sizes of 120, 180, and 240 months, respectively. The columns labeled 'RSk' and 'RWWD' represent the returns obtained from trading strategies based on the forecast model and the benchmark model, respectively. Asterisks (**, *, *) mark significance at the 1%, 5%, and 10% thresholds, respectively.

commodity currencies, regardless of the choice of window sizes and estimation schemes.

6.4. Comparison with common predictors

We further explore whether realized skewness of oil price returns contains more effective information than common predictors of crude oil markets, such as oil prices and oil volatility. We conduct the following univariate and bivariate regressions to compare the predictability of realized skewness and common predictors on exchange rate returns.

We use univariate regression to evaluate the predictive ability of a single predictor, which can be expressed as:

$$r_{i,t+1} = \alpha_i + \varphi_i X_t^k + \varepsilon_{i,t+1}, \quad (21)$$

where X_t^k denotes the changes in oil prices (Δp_t) or oil volatility (RV_t). The changes in oil prices are the first-order log differences in oil prices. And we use the RV of oil prices calculated in Eq.(2) as the oil volatility.[†]

Then we use the bivariate regression to investigate the predictive ability of realized skewness of oil price returns after accounting for common predictors.

$$r_{i,t+1} = \alpha_i + \varphi_i X_t^k + \beta_i RSk_t + \varepsilon_{i,t+1}. \quad (22)$$

Table 11 reports the regression results. Specifically, in Panel A, the univariate regression slope coefficients of Δp_t for all

currencies except for JPY and CHF are significant at the 5% level, with the largest R^2 value reaching 2.610% for CAD. However, in Panel B, the coefficients of RV_t are almost always insignificant and yield less than 0.5% R^2 . In contrast to the results in Table 2, the RSk_t generally exhibits stronger predictability for exchange rate returns, especially when compared with the RV_t .

Turning to the bivariate regressions, when the additional variables and RSk_t are combined to predict jointly, the coefficients of RSk_t for all currencies except for JPY remain statistically significant. According to Rapach *et al.* (2016), when two variables are integrated and their information does not overlap, forecasting performance can be enhanced. The R^2 s for all bivariate regressions are improved compared to the corresponding univariate regressions, with the highest value of 4.390% for NOK in Panel A. The R^2 s for all bivariate regressions are nearly equal to the total R^2 when realized skewness and common predictors are employed alone. The results indicate that the predictability provided by realized skewness is not influenced by common predictors like oil prices and oil volatility. Realized skewness offers supplementary predictive insights that go beyond the information already captured by these common predictors.

7. Conclusion

We propose a novel predictor based on the realized skewness of oil price returns. Using the exchange rate for 9 developed countries, we find that realized skewness can significantly predict exchange rate returns for 8 countries, except for JPY, both in- and out-of-sample. The economic analysis reveals

[†] We find similar results for implied volatility of CBOE Crude Oil ETF Volatility Index (OVX) which is obtained from <https://fred.stlouisfed.org/series/OVXCLS>, since its data are available only after 2007, we omit the results and they are available upon request.

Table 11. Comparison with other variables.

	Univariate			Bivariate				
	φ	t_{φ} -stat	R^2	β	t_{β} -stat	φ	t_{φ} -stat	R^2
Panel A: The change in oil prices (Δp_t)								
AUD	−0.031**	−2.256	1.313	−0.293***	−2.804	−0.021	−1.486	2.195
CAD	−0.027***	−2.959	2.610	−0.195***	−2.640	−0.020**	−2.104	3.665
CHF	−0.015	−1.253	0.310	−0.276**	−2.095	−0.006	−0.485	1.107
DKK	−0.026**	−2.041	1.081	−0.306***	−2.672	−0.015	−1.244	2.232
GBP	−0.036**	−2.394	2.198	−0.380***	−3.326	−0.023	−1.409	4.023
JPY	0.005	0.311	0.037	−0.248	−1.640	0.013	0.801	0.669
NOK	−0.040***	−2.763	2.368	−0.433***	−3.393	−0.026*	−1.761	4.390
NZD	−0.040***	−2.867	2.083	−0.245**	−2.092	−0.032**	−2.100	2.666
SEK	−0.031**	−2.066	1.370	−0.242**	−2.169	−0.023	−1.466	2.013
Panel B: The realized volatility of oil prices (RV_t)								
AUD	−0.035*	−1.705	0.307	−0.389***	−3.415	−0.045**	−2.321	2.021
CAD	−0.002	−0.147	0.002	−0.276***	−3.607	−0.008	−0.757	2.370
CHF	0.009	0.919	0.021	−0.292**	−2.202	0.002	0.194	1.022
DKK	0.003	0.257	0.003	−0.365***	−3.075	−0.006	−0.409	1.845
GBP	0.022	1.164	0.153	−0.462***	−4.460	0.011	0.576	3.199
JPY	−0.020	−0.960	0.104	−0.207	−1.347	−0.025	−1.113	0.602
NOK	−0.025	−1.372	0.161	−0.551***	−3.992	−0.038**	−2.180	3.867
NZD	0.001	0.053	0.000	−0.369***	−3.139	−0.008	−0.377	1.495
SEK	−0.020	−1.088	0.111	−0.342***	−2.971	−0.029	−1.552	1.558

The table reports the in-sample results using univariate and bivariate predictors to predict exchange rates. The univariate predictive model is represented by $r_{i,t+1} = \alpha_i + \varphi_i X_t^k + \varepsilon_{i,t+1}$, and the bivariate predictive model is $r_{i,t+1} = \alpha_i + \varphi_i X_t^k + \beta_i RSk_t + \varepsilon_{i,t+1}$. In Panels A and B, X_t^k denote the changes in oil price (Δp_t) or oil volatility (RV_t), respectively. The coefficient, the Newey and West (1987) t -statistics, and R^2 are shown in the table. The choice of bandwidth is based on a rule of thumb based on sample size. The detail is as follows: $m = \text{floor}(4 * (T/100) \wedge (2/9))$, where m is the bandwidth and T is the number of the whole sample. ***, **, and * indicate significance at the 1%, 5%, and 10% thresholds, respectively.

that, in comparison to the RWWD model, the trading strategy based on the predictive model can produce higher certainty equivalent return (CER), Sharpe ratios, and lower maximum drawdowns in 7 of the 9 countries' currencies, except for CHF and JPY.

Furthermore, we find that, compared to oil prices and volatility, realized skewness can provide complementary information. Our results are robust to alternative currencies, different window sizes, and rolling windows. Moreover, the predictive power of realized skewness for exchange rates stems from the jump and drift components.

More importantly, our findings highlight the economic mechanisms through which crude oil's realized skewness predicts exchange rates. Specifically, RSk influences exchange rates via three key channels: inflation rates, investor sentiment, and global risk. First, as an energy cost driver, crude oil skewness affects inflation, which in turn impacts exchange rates through purchasing power parity. Second, RSk captures shifts in investor sentiment, with positive skewness often signaling speculative demand or optimism, leading to currency overvaluation. Finally, RSk reflects changes in global risk appetite, which affects investors' demand for different risk premiums, thereby influencing currency returns.

Acknowledgments

We are grateful to Sadie Thrift (the editor), Aleksey Kolokolov (the associate editor), and two anonymous referees for their extremely helpful comments and suggestions. We thank seminar participants at Nanjing University of Science

and Technology for helpful comments and suggestions. This work is supported by the National Natural Science Foundation of China [72001110, 72371131].

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by National Natural Science Foundation of China [grant number: 72001110].

ORCID

Yaojie Zhang  <http://orcid.org/0000-0002-4220-1623>

References

Aleti, S., Bollerslev, T. and Siggaard, M., Intraday market return predictability culled from the factor zoo. *Manage. Sci.*, 2025, forthcoming.
Amaya, D., Christoffersen, P., Jacobs, K. and Vasquez, A., Does realized skewness predict the cross-section of equity returns? *J. Financ. Econ.*, 2015, **118**, 135–167.
Andersen, T. G. and Bollerslev, T., Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *Int. Econ. Rev. (Philadelphia)*, 1998, **39**, 885–905.

- Andersen, T. G., Bollerslev, T. and Diebold, F. X., Roughing it up: Including jump components in the measurement, modeling, and forecasting of return volatility. *Rev. Econ. Stat.*, 2007, **89**, 701–720.
- Andersen, T. G., Bollerslev, T., Diebold, F. X. and Labys, P., Modeling and forecasting realized volatility. *Econometrica*, 2003, **71**, 579–625.
- Baker, M. and Wurgler, J., Investor sentiment and the cross-section of stock returns. *J. Finance*, 2006, **61**, 1645–1680.
- Bali, T. G., Cakici, N. and Whitelaw, R. F., Maxing out: Stocks as lotteries and the cross-section of expected returns. *J. Financ. Econ.*, 2011, **99**, 427–446.
- Bandi, F. M., Kolokolov, A., Pirino, D. and Renò, R. Discontinuous trading in continuous-time econometrics. Available at SSRN 4351618, 2023.
- Bandi, F. M. and Russell, J. R., Microstructure noise, realized variance, and optimal sampling. *Rev. Econ. Stud.*, 2008, **75**, 339–369.
- Barsky, R. B. and Kilian, L., Oil and the macroeconomy since the 1970s. *J. Econ. Perspect.*, 2004, **18**, 115–134.
- Bollerslev, T. and Todorov, V., Tails, fears, and risk premia. *J. Finance*, 2011, **66**, 2165–2211.
- Bollerslev, T., Todorov, V. and Xu, L., Tail risk premia and return predictability. *J. Financ. Econ.*, 2015, **118**, 113–134.
- Breen, J. D. and Hu, L., The predictive content of oil price and volatility: New evidence on exchange rate forecasting. *J. Int. Financ. Mark. I*, 2021, **75**, 101454.
- Brunnermeier, M., Xia, K., Nagel, S., Pedersen, L. and Xia, H., Carry trades and currency crashes. *NBER. Macroecon. Annu.*, 2008, **23**, 313–348.
- Campbell, J. Y. and Thompson, S. B., Predicting excess stock returns out of sample: Can anything beat the historical average? *Rev. Financ. Stud.*, 2008, **21**, 1509–1531.
- Chen, S.-S., Oil price pass-through into inflation. *Energ. Econ.*, 2009, **31**, 126–133.
- Chen, S.-S. and Chen, H.-C., Oil prices and real exchange rates. *Energ. Econ.*, 2007, **29**, 390–404.
- Chen, Y.-c. and Rogoff, K., Commodity currencies. *J. Int. Econ.*, 2003, **60**, 133–160.
- Chen, Y.-C., Rogoff, K. S. and Rossi, B., Can exchange rates forecast commodity prices? *Q. J. Econ.*, 2010, **125**, 1145–1194.
- Choi, S., Furceri, D., Loungani, P., Mishra, S. and Poplawski-Ribeiro, M., Oil prices and inflation dynamics: Evidence from advanced and developing economies. *J. Int. Money. Finance.*, 2018, **82**, 71–96.
- Clark, T. E. and McCracken, M. W., Improving forecast accuracy by combining recursive and rolling forecasts. *Int. Econ. Rev. (Philadelphia)*, 2009, **50**, 363–395.
- Clark, T. E. and West, K. D., Approximately normal tests for equal predictive accuracy in nested models. *J. Econom.*, 2007, **138**, 291–311.
- Clifton Green, T. and Hwang, B.-H., Initial public offerings as lotteries: Skewness preference and first-day returns. *Manage. Sci.*, 2012, **58**, 432–444.
- Conflitti, C. and Luciani, M., Oil price pass-through into core inflation. *Energ. J.*, 2019, **40**, 221–247.
- Conrad, J., Dittmar, R. F. and Ghysels, E., Ex ante skewness and expected stock returns. *J. Finance*, 2013, **68**, 85–124.
- Conrad, J., Kapadia, N. and Xing, Y., Death and jackpot: Why do individual investors hold overpriced stocks? *J. Financ. Econ.*, 2014, **113**, 455–475.
- Corsi, F., Pirino, D. and Renò, R., Threshold bipower variation and the impact of jumps on volatility forecasting. *J. Econom.*, 2010, **159**, 276–288.
- Corte, P. D., Riddiough, S. J. and Sarno, L., Currency premia and global imbalances. *Rev. Financ. Stud.*, 2016, **29**, 2161–2193.
- Corte, P. D., Sarno, L. and Tsiakas, I., An economic evaluation of empirical exchange rate models. *Rev. Financ. Stud.*, 2009, **22**, 3491–3530.
- Cremers, M. and Weinbaum, D., Deviations from put-call parity and stock return predictability. *J. Financ. Quant. Anal.*, 2010, **45**, 335–367.
- David, A., Fluctuating confidence in stock markets: Implications for returns and volatility. *J. Financ. Quant. Anal.*, 1997, **32**, 427–462.
- Della Corte, P., Ramadorai, T. and Sarno, L., Volatility risk premia and exchange rate predictability. *J. Financ. Econ.*, 2016, **120**, 21–40.
- DeMiguel, V., Garlappi, L. and Uppal, R., Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Rev. Financ. Stud.*, 2009, **22**, 1915–1953.
- Fernandez-Perez, A., Frijns, B., Fuertes, A.-M. and Miffre, J., The skewness of commodity futures returns. *J. Bank. Financ.*, 2018, **86**, 143–158.
- Ferraro, D., Rogoff, K. and Rossi, B., Can oil prices forecast exchange rates? An empirical analysis of the relationship between commodity prices and exchange rates. *J. Int. Money. Finance.*, 2015, **54**, 116–141.
- Filippou, I., Rapach, D., Taylor, M. P. and Zhou, G., Out-of-sample exchange rate prediction: A machine learning perspective. Available at SSRN 3455713, 2023.
- Fong, W. M., Market-wide sentiment and market returns. *J. Asset Manage.*, 2015, **16**, 316–328.
- French, K. R., Schwert, G. W. and Stambaugh, R. F., Expected stock returns and volatility. *J. Financ. Econ.*, 1987, **19**, 3–29.
- Fukasawa, M. and Matsushita, K., Realized cumulants for martingales, 2021, 2021.
- Gao, L., Hitzemann, S., Shaliastovich, I. and Xu, L., Oil volatility risk. *J. Financ. Econ.*, 2022, **144**, 456–491.
- Han, L., Xu, Y., Zhang, Q. and Zhu, X., Oil strikes back: Trend factors and exchange rates. *J. Money Credit Bank.*, 2024, forthcoming.
- Harvey, C. R. and Siddique, A., Conditional skewness in asset pricing tests. *J. Finance*, 2000, **55**, 1263–1295.
- Hong, S. and Wei, X., Blockbuster or bust? silver screen effect and stock returns. *Rev. Financ.*, 2025, **29**, 603–632.
- Huang, D., Jiang, F., Tu, J. and Zhou, G., Investor sentiment aligned: A powerful predictor of stock returns. *Rev. Financ. Stud.*, 2015, **28**, 791–837.
- Iyke, B. N., Phan, D. H. B. and Narayan, P. K., Exchange rate return predictability in times of geopolitical risk. *Int. Rev. Financ. Anal.*, 2022, **81**, 102099.
- Jiang, F., Lee, J. A., Martin, X. and Zhou, G., Manager sentiment and stock returns. *J. Financ. Econ.*, 2019a, **132**, 126–149.
- Jiang, X., Han, L. and Yin, L., Can skewness of the futures-spot basis predict currency spot returns? *J. Futures Markets*, 2019b, **39**, 1435–1449.
- Jiang, X., Han, L. and Yin, L., Can skewness predict currency excess returns? *N. Am. J. Econ. Financ.*, 2019c, **48**, 628–641.
- Jobson, J. D. and Korkie, B. M., Performance hypothesis testing with the sharpe and treynor measures. *J. Finance*, 1981, **36**, 889–908.
- Jondeau, E., Zhang, Q. and Zhu, X., Average skewness matters. *J. Financ. Econ.*, 2019, **134**, 29–47.
- Kelly, B. and Jiang, H., Tail risk and asset prices. *Rev. Financ. Stud.*, 2014, **27**, 2841–2871.
- Kilian, L. and Zhou, X., The impact of rising oil prices on U.S. inflation and inflation expectations in 2020–23. *Energ. Econ.*, 2022, **113**, 106228.
- Kim, T.-H. and White, H., On more robust estimation of skewness and kurtosis. *Finance Research Letters*, 2004, **1**, 56–73.
- Kohlscheen, E., Avalos, F. H. and Schrimpf, A., When the walk is not random: Commodity prices and exchange rates. *Int. J. Cent. Bank.*, 2017, **13**, 121–158.
- Kolokolov, A., Estimating jump activity using multipower variation. *J. Bus. Econ. Stat.*, 2022, **40**, 128–140.
- Kraus, A. and Litzenberger, R. H., Skewness preference and the valuation of risk assets. *J. Finance*, 1976, **31**, 1085–1100.
- Li, J., Tsiakas, I. and Wang, W., Predicting exchange rates out of sample: Can economic fundamentals beat the random walk? *J. Financ. Economet.*, 2015, **13**, 293–341.
- Liao, J., Shi, Y. and Xu, X., Why is the correlation between crude oil prices and the US dollar exchange rate time-varying?—Explanations based on the role of key mediators. *Int. J. Financ. Stud.*, 2018, **6**, 61–73.

- Liu, L., Tan, S. and Wang, Y., Can commodity prices forecast exchange rates? *Energ. Econ.*, 2020, **87**, 104719.
- Liu, Y., Han, L. and Wu, Y., Can skewness predict CNY-CNH spread? *Financ. Res. Lett.*, 2022, **46**, 102392.
- Liu, Z., Wang, K. and Liu, J., Realized skewness at high frequency and the link to a conditional market premium. In *Proceedings of the Asian Finance Association (AsFA) 2013 Conference*, 2014.
- Lizardo, R. A. and Mollick, A. V., Oil price fluctuations and U.S. dollar exchange rates. *Energ. Econ.*, 2010, **32**, 399–408.
- Meese, R. and Rogoff, K., The out-of-sample failure of empirical exchange rate models: Sampling error or misspecification? In *Exchange Rates and International Macroeconomics*, edited by J. Frenkel, pp. 67–112, 1983a (University of Chicago Press: Chicago).
- Meese, R. A. and Rogoff, K., Empirical exchange rate models of the seventies: Do they fit out of sample? *J. Int. Econ.*, 1983b, **14**, 3–24.
- Mommel, C., Performance hypothesis testing with the sharpe ratio. *Financ. Lett.*, 2003, **1**, 21–23.
- Merton, R. C., On estimating the expected return on the market: An exploratory investigation. *J. Financ. Econ.*, 1980, **8**, 323–361.
- Mitton, T. and Vorkink, K., Equilibrium underdiversification and the preference for skewness. *Rev. Financ. Stud.*, 2007, **20**, 1255–1288.
- Musa, A., Salisu, A. A., Abulbashar, S. and Okoronkwo, C. D., Oil price uncertainty and real exchange rate in a global VAR framework: A note. *J. Econ. Finan.*, 2022, **46**, 704–712.
- Neely, C. J., Rapach, D. E., Tu, J. and Zhou, G., Forecasting the equity risk premium: The role of technical indicators. *Manage. Sci.*, 2014, **60**, 1772–1791.
- Neuberger, A., Realized skewness. *Rev. Financ. Stud.*, 2012, **25**, 3423–3455.
- Neuberger, A. and Payne, R., The skewness of the stock market over long horizons. *Rev. Financ. Stud.*, 2021, **34**, 1572–1616.
- Newey, W. K. and West, K. D., A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 1987, **55**, 703–708.
- Nguyen, B. H., Okimoto, T. and Tran, T. D., Uncertainty-dependent and sign-dependent effects of oil market shocks. *J. Commod. Mark.*, 2022, **26**, 100207.
- Patton, A. J. and Sheppard, K., Good volatility, bad volatility: Signed jumps and the persistence of volatility. *Rev. Econ. Stat.*, 2015, **97**, 683–697.
- Paye, B. S., Déjà vol': Predictive regressions for aggregate stock market volatility using macroeconomic variables. *J. Financ. Econ.*, 2012, **106**, 527–546.
- Rapach, D., Ringgenberg, M. C. and Zhou, G., Short interest and aggregate stock returns. *J. Financ. Econ.*, 2016, **121**, 46–65.
- Rapach, D., Strauss, J. K. and Zhou, G., Out-of-sample equity premium prediction: Combination forecasts and links to the real economy. *Rev. Financ. Stud.*, 2010, **23**, 821–862.
- Reboredo, J. C., Rivera-Castro, M. A. and Zebende, G. F., Oil and US dollar exchange rate dependence: A detrended cross-correlation approach. *Energ. Econ.*, 2014, **42**, 132–139.
- Ren, Y., Liang, X. and Wang, Q., Short-term exchange rate forecasting: A panel combination approach. *J. Int. Financ. Mark. I*, 2021, **73**, 101367.
- Rossi, B., Exchange rate predictability. *J. Econ. Lit.*, 2013, **51**, 1063–1119.
- Salisu, A. A., Adekunle, W., Alimi, W. A. and Emmanuel, Z., Predicting exchange rate with commodity prices: New evidence from westerlund and narayan (2015) estimator with structural breaks and asymmetries. *Resour. Policy*, 2019, **62**, 33–56.
- Sarno, L., Schneider, P. and Wagner, C., Properties of foreign exchange risk premiums. *J. Financ. Econ.*, 2012, **105**, 279–310.
- Schmeling, M., Investor sentiment and stock returns: Some international evidence. *J. Empir. Financ.*, 2009, **16**, 394–408.
- Schwert, G. W., Why does stock market volatility change over time? *J. Finance*, 1989, **44**, 1115–1153.
- Sek, S. K., Teo, X. Q. and Wong, Y. N., A comparative study on the effects of oil price changes on inflation. *Proc. Econ. Financ.*, 2015, **26**, 630–636.
- Śmiech, S., Papież, M., Rubaszek, M. and Snarska, M., The role of oil price uncertainty shocks on oil-exporting countries. *Energ. Econ.*, 2021, **93**, 105028.
- Stambaugh, R. F., Yu, J. and Yuan, Y., The short of it: Investor sentiment and anomalies. *J. Financ. Econ.*, 2012, **104**, 288–302.
- Yin, L. and Feng, J., Oil market uncertainty and international business cycle dynamics. *Energ. Econ.*, 2019, **81**, 728–740.
- Yin, L., Su, Z. and Lu, M., Is oil risk important for commodity-related currency returns? *Res. Int. Bus. Financ.*, 2022, **60**, 101604.
- Zhang, Z., He, M., Zhang, Y. and Wang, Y., Realized skewness and the short-term predictability for aggregate stock market volatility. *Econ. Model.*, 2021, **103**, 105614.